

Assignment-CO1, CO2, CO3

CO-1 Questions

1. Reduce the following matrices to *Echelon* form and *Row-Reduced Echelon* form, and hence determine the rank:

$$A = \begin{bmatrix} 10 & 8 & 6 & 4 \\ 2 & 0 & -2 & -4 \\ -6 & -8 & -10 & -12 \\ -2 & -4 & -6 & -8 \end{bmatrix}$$

2. Test for the consistency and if possible, solve the following system of equations:

$$\begin{aligned} x - y + 2z &= 2, \\ 2x + y + 4z &= 7, \\ 4x - y + z &= 4 \end{aligned}$$

3. Let $A = \begin{bmatrix} 1 & -1 & 1 \\ 1 & 1 & 1 \\ 2 & 3 & \alpha \end{bmatrix}$ and $b = \begin{bmatrix} 1 \\ 3 \\ \beta \end{bmatrix}$. Then for what values of α and β , the system

$AX = b$ has (i) no solution (ii) unique solution (iii) infinitely many solutions.

4. For what values of $\lambda \in R$, the following system of equations has (i) no solution, (ii) a unique solution, and (iii) infinitely many solutions ?

$$\begin{aligned} (5 - \lambda)x + 4y + 2z &= 4, \\ 4x + (5 - \lambda)y + 2z &= 4, \\ 2x + 2y + (2 - \lambda)z &= 2. \end{aligned}$$

5. Determine all 2×2 matrices A such that A has eigen values 2 and -1 with corresponding eigenvectors $[1 \ 0]^T$ and $[2 \ 1]^T$, respectively.

6. Determine the eigenvalues and eigenvectors of the matrix:

$$A = \begin{bmatrix} 2 & -2 & -2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$$

7. Let A be a 3×3 matrix with real entries such that $\det(A - I) = 0$. If the $\text{trace}(A) = 13$ and $\det(A) = 32$, then find the sum of squares of the eigenvalues of A .

8. Applying the Gauss-Elimination method, solve the following system:

$$\begin{aligned} 2x_1 - x_2 + 2x_3 + 2x_4 &= 14 \\ x_1 + 2x_2 - x_3 + x_4 &= 6 \\ -x_1 + x_2 + 2x_3 - x_4 &= 3 \\ x_1 + x_2 - x_3 + 2x_4 &= 8 \end{aligned}$$

9. Applying Gauss-Jordan Method, solve the following system of equations

$$\begin{aligned}x_2 + 2x_3 &= 3 \\x_1 + 2x_2 + 4x_3 + 2x_4 &= 2 \\x_1 + 3x_2 + 6x_3 + x_4 &= 5 \\2x_1 + 4x_2 + 8x_3 + 2x_4 &= 4\end{aligned}$$

10. Determine the inverse of the following matrices using Gauss-Jordan method:

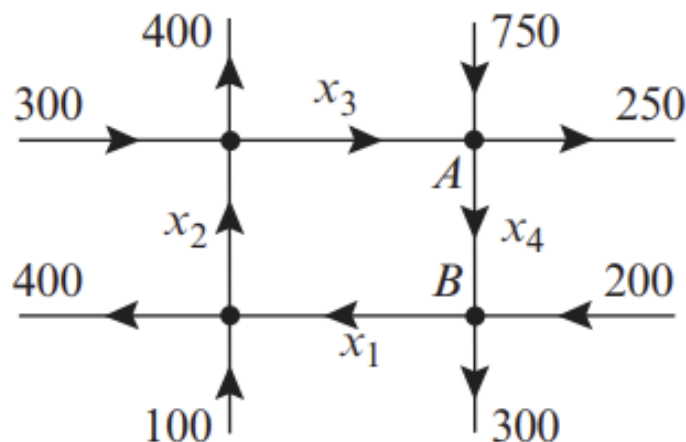
$$\begin{aligned}\text{(a)} \quad A &= \begin{bmatrix} 2 & 1 & -1 \\ 5 & 2 & -3 \\ 0 & 2 & 1 \end{bmatrix} \\ \text{(b)} \quad B &= \begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & -2 \\ -3 & 4 & -4 \end{bmatrix}\end{aligned}$$

11. Let L , D and U be a lower triangular matrix, diagonal matrix, and the upper triangular matrix, respectively. Decompose the following matrix

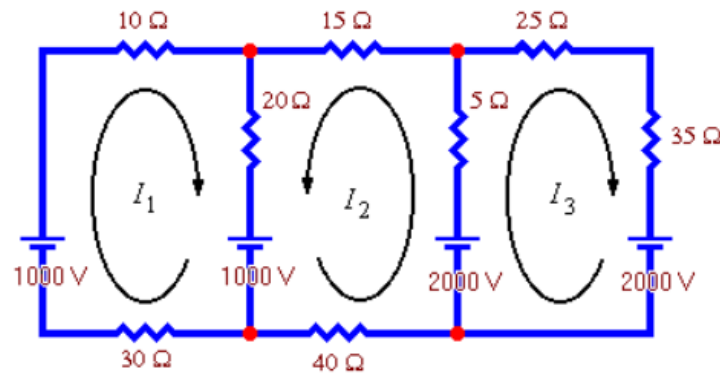
$$A = \begin{bmatrix} 1 & 3 & 4 \\ 3 & 15 & 20 \\ 2 & 18 & 26 \end{bmatrix}$$

into a product of L , D and U .

12. The accompanying figure shows a network of one-way streets with traffic flowing in the directions indicated. The flow rates along the streets are measured as the average number of vehicles per hour.
- Set up a linear system whose solution provides the unknown flow rates.
 - Solve the system for the unknown flow rates.
 - If the flow along the road from A to B must be reduced for construction, what is the minimum flow that is required to keep traffic flowing on all roads?



13. Determine the current i_1 , i_2 , i_3 in the following electrical circuit using any matrix method.



Answer. $i_1 = -4.56$, $i_2 = 13.7$, $i_3 = -1.05$.

14. Associate the numbers with their corresponding letters from the following table

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
P	Q	R	S	T	U	V	W	X	Y	Z	Blank	?	!	
↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕
15	16	17	18	19	20	21	22	23	24	25	26	27	28	

- a) Encode the following Message “HAPPY FAMILY” by using key matrix

$$\begin{bmatrix} -3 & -3 & -4 \\ 0 & 1 & 1 \\ 4 & 3 & 4 \end{bmatrix}$$

- b) Decrypt the following Message “GOOD MORNING TO ALL”. By using key

matrix $A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$.

CO-2 Questions

15. Let V be the set of all pairs (x, y) of real numbers, and let F be the field of real numbers. Define

- (i) $(x, y) + (x_1, y_1) = (x + y_1, y + x_1)$
 $c(x, y) = (cx, cy)$
(ii) $(x, y) + (x_1, y_1) = (0, 0)$
 $c(x, y) = (cx, cy)$

Is V , with these operations, a vector space over the field of real numbers?

16. Determine whether the set W is a subspace of $\mathbb{R}^3$ with the standard operations. Justify your answer.

- (a) $W = \{(s, t, s + t) : s \text{ and } t \text{ are real numbers}\}$
(b) $W = \{(x_1, 1/x_1, x_3) : x_1 \text{ and } x_3 \text{ are real numbers, } x_1 \neq 0\}$.

17. Let V be the set of all matrices of the form $\begin{pmatrix} 1 & p \\ q & 1 \end{pmatrix}$ where $p, q \in R$. Determine whether or not the set V is a vector space under standard addition and scalar multiplication of matrices.

18. Let $V = R^2$ and define $\begin{pmatrix} a \\ b \end{pmatrix} + \begin{pmatrix} c \\ d \end{pmatrix} = \begin{pmatrix} a + 2c \\ b + 2d \end{pmatrix}$ and $k \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} ka \\ kb \end{pmatrix}$.

Determine whether V is a vector space. Also find additive identity and additive inverse.

19. Verify that the set $M_{2 \times 2}$ of 2×2 matrices with real number entries is a vector space under the natural entry-by-entry operations:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} + \begin{pmatrix} w & x \\ y & z \end{pmatrix} = \begin{pmatrix} a + w & b + x \\ c + y & d + z \end{pmatrix} \text{ and } k \cdot \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} ka & kb \\ kc & kd \end{pmatrix}.$$

20. Show that the subset $W = \{A \in M_n(R) \mid \text{Trace}(A) = 0\}$ is a subspace of vector space $M_n(R)$.

21. Let V be the vector space of all 2×2 matrices over the field R . Show that W is not a subspace of V , where W contains all 2×2 matrices with zero determinants.

22. Check whether the following vectors are linearly independent or dependent?

$$v_1 = \begin{pmatrix} 3 \\ 4 \\ 5 \end{pmatrix}, v_2 = \begin{pmatrix} 2 \\ 9 \\ 2 \end{pmatrix} \text{ and } v_3 = \begin{pmatrix} 4 \\ 18 \\ 4 \end{pmatrix}$$

23. Determine whether the set $\{x^3 - x, 2x^2 + 4, -2x^3 + 3x^2 + 2x + 6\}$ is linearly independent or linearly dependent in $P_3(\mathbb{R})$.

24. Find a basis of $W_1 \cap W_2$, where W_1 and W_2 are the subspaces of R^4 generated by $(1, 1, 1, 1), (1, 1, -1, -1), (1, -1, 1, -1)$ and $(1, 1, 0, 0), (1, -1, 1, 0), (2, -1, 1, -1)$, respectively.

25. Show that the vectors $\alpha = (1, 0, -1), \beta = (1, 2, 1), \gamma = (0, -3, 2)$, form a basis for R^3 . Express each of the standard basis vectors as a linear combination of α, β, γ .

26. Find the row rank, column rank, rank, and nullity of the matrix

$$\begin{bmatrix} 1 & 2 & 0 & 2 & 1 \\ -1 & -2 & 1 & 1 & 0 \\ 1 & 2 & -3 & -7 & 2 \\ 1 & 2 & -2 & -4 & 3 \end{bmatrix}$$

27. Find a basis for row space, column space, and null space for the following matrix:

$$M = \begin{bmatrix} 1 & 2 & 0 & 2 & 5 \\ -2 & -5 & 1 & -1 & 8 \\ 0 & -3 & 3 & 4 & 1 \\ 3 & 6 & 0 & -7 & 2 \end{bmatrix}$$

CO-3 Questions

1. Determine whether the function is a linear transformation.
 - (a) $T: R^2 \rightarrow R^2, T(x, y) = (x, 1)$
 - (b) $T: R^2 \rightarrow R^2, T(x, y) = (x, y^2)$
 - (c) $T: R^3 \rightarrow R^3, T(x, y, z) = (x + y, x - y, z)$

2. For the linear transformations $T: R^3 \rightarrow R^3$, determine $T(x, y, z)$.
 - (a) $T(2, 0, 2) = (2, 4, -1), T(4, 2, 0) = (1, 3, -2), T(0, 2, 5) = (0, -2, 2)$

3. For the linear transformation $T: R^2 \rightarrow R^2$ given by $A = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$ such that $T(12, 5) = (13, 0)$.

4. Define the linear transformation T by $T(v) = Av$. Find (i) $\ker(T)$ or $\text{null}(T)$, (ii) $\text{nullity}(T)$, (iii) $\text{range}(T)$ or $\text{Im}(T)$, and (iv) $\text{rank}(T)$.
 - (a) $A = \begin{bmatrix} 2 & 0 \\ 2 & 4 \end{bmatrix}$
 - (b) $A = \begin{bmatrix} 5 & -3 \\ 1 & 1 \\ 1 & -1 \end{bmatrix}$

5. If $T: R^3 \rightarrow R^3$ is defined by $T(x, y, z) = (2x - z, 3x - 2y, x - 2y + z)$,
 - (a) Determine the null space and nullity of T
 - (b) Determine whether T is one-one and onto
 - (c) Find a basis for null space of T
 - (d) Determine the range space or image space and rank of T
 - (e) Find a basis of range space of T